

Improving Environmental Knowledge with a Serious Game: An experimental study

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Figure 1: Eco-City Builder is a serious game developed to strengthen players' environmental knowledge. A thriving city has to be built from the ground up. Players are encouraged to increase their economies while reducing environmentally harmful behavior.

ABSTRACT

Climate change mitigation is the century's obligation to humanity. People all throughout the world have too little knowledge and comprehension of the planet's ecosystem. Increasing people's awareness and knowledge about the environment might help reduce environmental-hostile behavior. We propose a serious game that helps players to learn more about the environment in a fun and engaging manner. The game's effectiveness in teaching players about the environment is evaluated, and its role as a mediation tool is estimated by measuring environmental knowledge and consciousness using literature-based questionnaires before and after the game is played. Our results show significant differences in participants' environmental knowledge. Differences in environmental consciousness are not significant. Strengths and weaknesses of the developed serious game based on user feedback are identified to improve future iterations.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; Visualization design and evaluation methods.

KEYWORDS

serious games, environmental knowledge, pre- and post-test

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1 INTRODUCTION

The century's responsibility to humanity is to mitigate the impacts of climate change. People tend to have too little knowledge about the environment, as is shown by studies all over the world: 11th grade students from New York show too little knowledge of the environment [28]. In the Dominican Republic, 12th grade students did not know enough about their local environment in their own country [24]. Besides these objective measurements, perceived satisfaction with the German people's own actions for protecting the environment saw a decrease of 17% from 2016 to 2018 [29]. Another publication by DeChano also indicates low scores for environmental knowledge across multiple countries and states that this needs to be addressed [7]. The United Nations Educational, Scientific, and Cultural Organization (UNESCO) even defined *Quality Education for Sustainable Development* as one of their main goals [23]. In order to address the aforementioned situation, we propose our developed serious game as a mediation tool for environmental knowledge. Our statistical analysis with qualitative input on the game's merits and faults, which we refer to as its effectiveness, indicates that the serious game functions as a medium for knowledge mediation.

2 BACKGROUND

The effectiveness of learning using serious games is subject of debate, and receptions on this topic are generally mixed [11, 12, 22, 30].

Flowers and Aggarwal, for example, advocate for the serious game *Second Life*. Their report takes place in a hospital, and it uses a digital version of the hospital as a learning tool so that new hospital workers can get a feel for what it's like to work there. The

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utility of their tool is thought to be beneficial by trained medical professionals [10].

Starker et al. present another theoretical framework that uses a location-based augmented reality app for nature trails to encourage people to think about the environment in a sustainable way. They thoroughly integrate their game design framework into the literature and provide additional key aspects for training sustainability skills. However, there are no objective measures provided for the application's effectiveness [26].

Authors present a distinct quantity of serious games to promote various environmental topics. Some games emphasize how important it is to recycle correctly [15, 16], while others help players get back in touch with how food is made and how many resources it takes [27].

Despite the commendable development efforts and the deeply literature-rooted decisions made during their implementation, the effectiveness of each serious game should be proven as objectively as possible on a case-by-case basis. Madani et al. mention the urgency of publishing more research on the efficacy of serious games [19]. The literature review about experimental serious game studies by Nguyen and Hallinger advises that future research needs to employ statistically appropriate tools and represent them in decent detail [21].

3 METHODS

The developed serious game allows players to build a prosperous city while minimizing their negative environmental impact. We conducted a controlled experiment that captured the environmental knowledge and consciousness of participants using questionnaires at two different points in time. There was a one-month break in between the recordings. Before the second recording, participants played the developed serious game for at least half an hour before answering the second questionnaire. Players also provide information about their playing routine, their self-assessed knowledge of the environment, and their attitude towards the environment. The research question guiding this experiment is: Does the provided serious game effectively work as a mediation tool for environmental knowledge and consciousness?

3.1 System design

We developed a city-building game named *Eco-City Builder* where players must expand their city and supply their citizens with food and electricity (Figure 1). The game features the concept of a non-renewable currency for the city's ecology (*eco-points*). It is consumed for almost any action in the game. The goal of the game is to reach a certain population at both population levels and maintain a certain amount of happiness among the citizens while not fully depleting the available eco-points. The challenge is that players only have a limited amount of eco-points that they can spend on environmentally harmful actions.

Figure 2 (left) shows that putting up and running a building costs money and uses up eco-points over and over again. New buildings can be unlocked by achieving certain in-game goals. Players are incentivized to increase their experience level in order to unlock more potentially environmentally friendly buildings. This is due to having only environmentally hostile production facilities at the

beginning of the game. There's a fine line between economic growth and irreversible destruction of the environment. If the eco-bar is depleted or the playtime of 30 minutes ends, the game is over.

Achievability and meaningfulness are psychological dimensions that are thought to be underrepresented in recent environmental games [9]. Therefore, the core gameplay of *Eco-City Builder* revolves around making meaningful choices between economic growth and sustainability. Most placeable objects can be upgraded to increase, for instance, the production of power in order to meet the game's objectives. But every upgrade comes with a price: upgrades cost money or energy and their consumption of eco-points may also increase or decrease dramatically.

The penalty for not having enough energy or food is large: in the game, citizens will move out of their buildings, production of food stops, and a player's income decreases significantly. Players get immediate feedback and are penalized for having too little energy, food, and happiness. Putting houses, apartments, or skyscrapers next to production facilities also makes people less happy. Putting parks or trees next to housing infrastructure, on the other hand, makes people happier.

But transitioning to renewable energy production is not that easy. Wind for wind turbines is simulated by real weather data. Also, a dynamic day and night cycle removes the ability for solar parks to generate electricity during the night, when the sun does not shine. Upgrades, like batteries, can only account for a small amount of the energy required to fulfill the game's goals. There are certain thresholds at which ecological catastrophes occur, which we consider peak gameplay moments. Players might remember these catastrophes better since they are the game's climax, and peak-end effects tend to have an influence on the retrospective perception of players [13]. When inevitably reaching one of the three climate catastrophes, the game's visuals change. The background music changes to a more darkened tone, guiding the players' emotions toward the intent of making this moment feel bad [17]. The game's ecological crises can be seen in Figure 3. They range from a drought to polluted sea water and global warming accompanied by floods. Besides the game's plentiful written explanations for certain placeable objects and upgrades, these crises change how the player's cities look in a drastic way. The water level rises and falls; dead fish lie in the now polluted water; and the green grass has turned into a dried-out wasteland.

Ang contends that gameplay and story should not compete but collaborate to give the player a more interesting experience [1]. Therefore, we decided to highlight the player's actions prior to the conclusion of the game. A cutscene based on the player's use of eco-points plays, giving players a chance to reflect about their choices.

3.2 Experimental Design

The research design is a Within-Subjects design, where participants were asked to conscientiously fill out the first questionnaire online and, after a one-month break in between, to come in person. At the second appointment, after giving consent, participants played the city-building game and filled out the second questionnaire afterwards. When the game is over, the questionnaire is opened by



Figure 2: Gameplay screenshots. Left: All hideable information panels (goals, building information, tooltips) are opened. The game includes a range of useful written information about environmental and gameplay-related topics. Right: The player places a wind turbine next to a coal-fired and nuclear power plant. This snippet demonstrates that, while transitioning to renewable energy production, it may be necessary to continue using environmentally harmful energy production.



Figure 3: Environmental catastrophes in the serious game. Catastrophes occur when exceeding certain eco-point thresholds. Left: Falling water level, drought. Middle: Polluted water, dying fish. Right: Global warming, including floods.

pressing a button. Further, both questionnaires start with questions about environmental knowledge and consciousness, ensuring fewer signs of fatigue [18]. Demographic data about the sample population is also gathered, as described below. In addition, players were asked to answer further questions orally, including to what extent sustainability affects them in their daily lives and, in their opinion, what needs to be done in order to make people behave more environmentally friendly. In this study, the environmental knowledge questionnaire is retrieved from DeChano [7]. Our version did undergo small changes, which can be found in Table 6. Every environmental knowledge question has four possible answers, of which only one is correct. Following is the questionnaire about participants' pro-ecological world view, which we refer to as *environmental consciousness*. The New Environmental Paradigm (NEP) scale by Dunlap et al. [8] was used for this purpose, which can be found in Table 7. It features 15 questions on a 5-point Likert scale, with equitable amounts of pro- and anti-NEP items. The study was conducted from October 2022 until December 2022. Both questionnaires were translated into German. To maintain a correct translation, the German translation of the NEP questionnaire by Schleyer-Lindemann et al. [25] was used.

3.3 Sample population

33 players took part in the study (9 female, 24 male), see Table 1. Players earned mandatory *study points* for first-semester university completion. Players who only completed the online questionnaire were excluded from the data set. We also asked players about their frequency of playing computer games, with four players playing every day or every other day. Four are said to play on a yearly basis, while eight play monthly, seven weekly, and 10 daily, as can be seen in Table 1.

4 RESULTS

We calculated group comparisons for both the knowledge and awareness findings in order to make an educated prediction regarding the role of the serious game as a mediation instrument for environmental knowledge and consciousness. Furthermore, qualitative analysis based on orally posed questions was carried out.

4.1 Environmental knowledge

The results of the environmental knowledge can be seen in Table 2. The table demonstrates that playing the serious game increases environmental knowledge. In order to find out if the increase in knowledge of 9.61% from 70.91% before playing the developed serious game to 81.52% after playing the serious game is significant, paired t-tests were performed. These t-tests have preconditions: The residuals need to be normally distributed. We therefore calculated

gender	age			playing routine					
	15-19	20-24	25-29	never	yearly	monthly	weekly	alternate days	daily
Male	10	10	4	0	2	6	5	2	9
Female	2	5	2	2	2	2	2	0	1
n	12	15	6	2	4	8	7	2	10
%	36.36	45.46	18.18	6.1	12.12	24.24	21.21	6.1	30.1

Table 1: Demographics of the study's sample population. All participants were students and below their third semester. No participant identified as anything other than male or female.

a Shapiro-Wilk normality test with $W = 0.966$, $p\text{-value} = 0.384$. This test's null hypothesis assumes a normal distribution. With the large p -value above the significance level of $\alpha = 0.05$, it is not possible to reject the null hypothesis. The plots in Figure 4 depict this well. The environmental knowledge answers before and after playing the serious game are assumed to be normally distributed. Further, the y -variable needs to be metric, which is the case, and the two samples have to be interdependent, which is also the case. The results of the paired t -tests can be found in Table 3.

The first t -test is a parametric t -test, while the second t -test is one-sided, assuming an increase in environmental knowledge. Compared to before playing the serious game ($m = 70.91$, $sd = 11.49$), players experienced a significant increase in their environmental knowledge ($m = 81.52$, $sd = 11.96$), $t(32) = -3.71$; $p < 0.05$, $d = 0.646$. Cohen suggests that this difference is of mediocre effect size [5].

4.2 Environmental consciousness

The results of the environmental consciousness can be found in Table 4. There is a slight decrease in environmental consciousness after participants played the serious game. Overall, even if the average NEP scores are almost identical, the distributions seemed to be unfamiliar, as can be seen in Table 4. A small left-shift can be interpreted from before to after playing the serious game.

We examined whether there are significant differences for people with lower (*NEP-Low*) or higher (*NEP-High*) levels of environmental consciousness in the following analysis. We followed Ibanez and Roussel's instruction to create dummy variables [14]. This approach utilizes the Johnson-Neyman technique [4]. A score lower than four means being part of the *NEP-Low* group, while a score equal to or higher than four is considered being part of the *NEP-High* group. The distribution of these newly created groups for the answers given to the NEP scale is $NEP\text{-}Low_{Before} = 66.45\%$, $NEP\text{-}High_{Before} = 33.54\%$, $NEP\text{-}Low_{After} = 66.26\%$ and $NEP\text{-}High_{After} = 33.74\%$ (*Before* and *After* referring to before or after playing the serious game).

To find out if the differences between the regular NEP score, the *NEP-Low* score, and the *NEP-High* score before and after playing the serious game are significant, we calculated the rank sum equality of population tests. The results can be found in Table 5. The Mann-Whitney-Wilcoxon rank sum equality of population test was chosen because the NEP-scale is ordinal and the samples are dependent. As can be seen in Table 5, all of the calculated p -values are above the significance level of $\alpha = 0.05$. P -values: $NEP = 0.22$, $NEP\text{-}Low = 0.36$, $NEP\text{-}High = 0.94$. This means that there are no significant

differences before and after playing the serious game in the NEP scores, the *NEP-Low* scores, and the *NEP-High* scores.

4.3 Qualitative analysis

When players were asked orally to what extent sustainability affects them in their daily lives, most referred to recycling trash and buying fair-trade products. They also mentioned that it is difficult to buy fairly traded products due to their limited monthly income. They do, however, think that buying fair-trade products and reducing meat consumption are great ways to mitigate climate change. One player stated that *"the economy does not take ecology into account"*. Another player stated that there are *"double moral standards: When it comes to animal products, I'm not the one who would pay twice to buy an ecologically correct piece of meat"*.

When asked what they think needs to be done to make people behave more sustainable, players said that forcing people to act more environmentally friendly might be a possible solution to environmental problems. *Spoons and straws, for example, are forced to be made out of paper instead of plastic by the local government* one player stated. Adding such forced conditions is thought to be a useful climate mitigation technique.

One player stated that after playing the game, he or she now has a bad conscience. Another player also mentioned wanting to play another round, knowing what impact the available eco-points in the game have on the environment. The wish was expressed for a faster transition to renewable energies in the game, slowing the rate of growth of their city during the transition from coal-fired energy to wind and solar.

5 DISCUSSION

We compared participants' environmental knowledge and consciousness before and after playing a serious game developed to strengthen participants' environmental awareness. After playing the serious game, participants significantly increased their environmental knowledge with a moderate effect size. The game's effectiveness in educating players about the environment is high; however, environmental awareness decreased slightly, necessitating additional revisions.

According to Coyle, everybody should know the answers to the first 12 environmental knowledge questions used in this study [6]. The National Environmental Education and Training Foundation (NEETF) later set the criteria for competence regarding the environmental knowledge questionnaire at 75% correct answers [7]. Though Coyle's report is based in the United States, the problems of the environmental knowledge questions used in this study change very little in other western countries. In this study, the significant

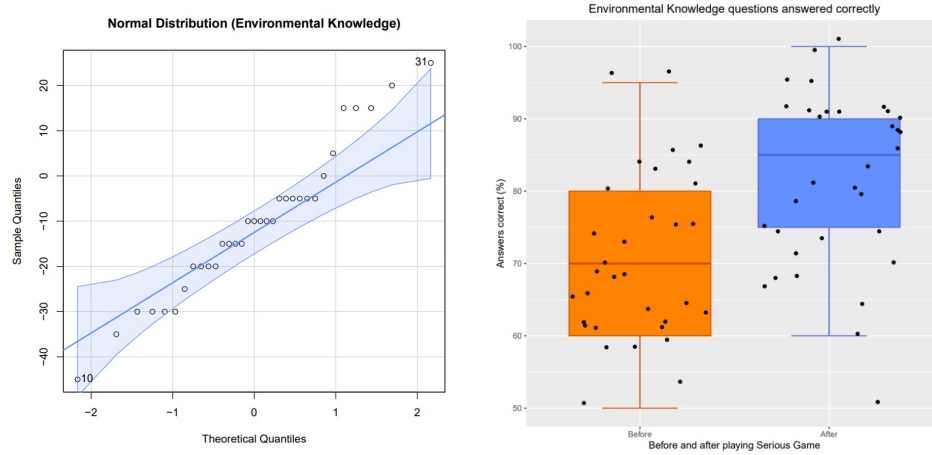


Figure 4: Normal distribution of the environmental knowledge answers. Left: Sample versus theoretical quantiles. Right: Before (orange) and after (blue) playing the serious game.

Knowledge	Before playing serious game	After playing serious game
1	100	100
2	94	91
3	94	85
4	42	73
5	88	70
6	73	79
7	58	73
8	85	91
9	94	85
10	73	85
11	55	79
12	85	91
13	18	79
14	64	85
15	85	82
16	27	58
17	85	88
18	70	64
19	85	97
20	45	79
Mean	71.91 (11.49)	81.52 (11.96)

Table 2: Percentages of participants who correctly answered each environmental knowledge question before and after playing the serious game. Standard deviations (SD) in parentheses.

t-test (paired)	t	df	p-value	confidence interval	cohens d
two-sided	-3.708	32	0.0007886	[-16.432 -4.780]	0.646
one-sided			0.0003943	[-Inf -5.762]	

Table 3: Paired t-test comparisons between the correct answered environmental knowledge questions for the measurement times before and after playing the serious game. The first t-test is a parametric, two-sided test. The second t-test is a parametric one-sided test with a lesser alternative hypothesis.

NEP item	Before playing serious game	After playing serious game
1	2.39	1.82
2	3.24	3.67
3	1.70	1.79
4	3.30	3.12
5	1.64	1.39
6	1.91	1.97
7	1.91	1.39
8	4.06	4.24
9	1.82	1.79
10	4.48	4.52
11	3.03	2.30
12	4.33	4.52
13	1.82	1.64
14	3.39	3.85
15	1.55	1.48
Mean	2.71	2.63

Table 4: Mean scores of participants on each NEP item before and after playing the serious game.

	mean scores		
	NEP	NEP-Low	NEP-High
Before	2.71	1.81	4.69
After	2.63	1.70	4.46
Mann-Whitney-Wilcoxon equality of population test (p-value)	0.22	0.3626	0.9351

Table 5: Quota of NEP-Low (value<4) and NEP-High (value>=4) participants before and after playing the serious game in percent. The population is compared for equality using a non-parametric Mann-Whitney-Wilcoxon test.

increase to an average of 81.52% correct answers to the environmental knowledge questions after playing the serious game exceeded the NEETF's criteria by 6.52%.

6 FUTURE WORK

Some players asked the experimenter gameplay-related questions during their play sessions. This might mean that the 30 minutes of maximum gameplay are too short to understand all gameplay mechanics. Changing the in-game tutorial and making the games' systems less complicated might help people understand them better, shifting the focus to a more comprehensible experience. Yun et al. stated that a game adapting to varying skill levels with a profile-based system might lead to an enhanced gaming experience [31]. In addition, future iterations of this study may need to involve a broader audience, including people with lower literacy levels and a wider range of skills.

Because repetition is an important part in learning [3], by removing the produced serious game from a controlled experimental context and encouraging replayability, players who return to play the game may be able to cultivate more environmental information. Further, players facing problems with the game's complexity could benefit from repetition as well. A reduction of the game's complexity to allow a better focus on the contents at hand is advised.

Since participants' environmental consciousness did not change significantly, and since changing people's behavior is a debatable

technique in HCI [2], the game might need further literature-based adjustments to make it more likely that participants' attitudes will become more environmentally friendly. To accomplish this, a deeply rooted implementation in the game of some of the intervention functions presented in Michie et al.'s *Behaviour Change Wheel* [20] could be possible. Persuasion, as one of these intervention functions, might be addressable by making renewable production facilities and upgrades within the game more distinguishable from their more environmentally hostile counterparts.

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A ENVIRONMENTAL KNOWLEDGE QUESTIONNAIRE

No.	<i>Environmental Knowledge Questions</i>
1.	The burning of fossil fuels has increased the carbon dioxide content of the atmosphere. What is the most immediate effect that this increasing amount of carbon dioxide is likely to have on our planet? a. warmer climate* b. cooler climate c. decreased relative humidity d. increased relative humidity
2.	Most elephants are killed every year to provide people with: a. trophies b. ivory* c. meat d. oil
3.	Burning coal for energy is a problem because it: a. releases carbon dioxide and other pollutants* b. decreases needed acid rain c. reduces the amount of ozone in the stratosphere d. is in limited supply
4.	Phosphates are harmful in sea water because they: a. can cause cancer in fish b. stop reproduction in fish c. make the water cloudy d. suffocate fish by increasing algae*
5.	The main source of pollution our surface water is caused by: a. dams on rivers b. chemical runoff from farms* c. methane gas d. water for sewage treatment plants
6.	Ecology is the study of the relationship between: a. different species of animals b. plants and the atmosphere c. organisms and their environments* d. man and the other animals
7.	Which of the following has the greatest impact on the earth's environment? a. damming rivers b. overpopulation* c. severe weather d. nuclear and power plants
8.	Precycling means that: a. people buy things that can be used again* b. more people should ride bicycles c. small children should wear the clothes of their older brother or sister d. items should be tested before we buy them
9.	Animals alive today are most likely to become extinct because: a. natural selection kills weaker animals b. where they live is getting too warm c. they are unable to reproduce because of pollution d. the habitat where they live is destroyed*
10.	Building a dam on a river can be harmful because it: a. makes the river muddy b. increases level of pollution in water c. causes the river to flood d. damages the river's natural ecosystem*
11.	An item which cannot be recycled and used again is: a. disposable diaper* b. newspaper c. tire d. plastic bottle
12.	What is the approximate human population of the earth? a. 6 billion b. 8 billion* c. 10 billion d. 12 billion
13.	Worldwide, most childhood deaths are the result of: a. starvation b. car accidents c. water pollution* d. child labour
14.	Which action can have the greatest impact on reducing the threat of global warming? a. recycling b. reducing energy use* c. composting d. planting a tree

No.	<i>Environmental Knowledge Questions</i>
15.	There are many different kinds of animals and plants, and they live in many different types of environments. What word is used to describe this idea? a. multiplicity b. biodiversity* c. socio-economics d. evolution
16.	The primary environmental benefit of a wetland is: a. a place to grow wild rice b. a hideout for ducks c. clean drinking water d. filtering water*
17.	Having ozone in the earth's upper atmosphere protects us from: a. alien invasion b. harmful, cancer-causing light rays* c. extremely cold winters d. extremely hot summers
18.	What is the name of the global agency that works to protect the physical earth? a. United National Environmental Programmes (UNEP)* b. United Nations Educational, Scientific and Cultural Organization (UNESCO) c. World Health Organization (WHO) d. United Nations Children's Fund (UNICEF)
19.	Carbon dioxide, methane, water vapour and nitrous oxide are examples of what? a. greenhouse gases* b. major atmospheric components c. major gases found in car exhaust d. are transpired by plants
20.	Approximately 70% of all fresh water withdrawn for human use is used for a. drinking b. cooking c. washing people and clothing d. irrigation*

Table 6: Environmental Knowledge questionnaire based upon the original work of [7]. The answers highlighted with a * are correct. The answers to the 12th question have been updated to suit the current global scenario.

B ENVIRONMENTAL CONSCIOUSNESS QUESTIONNAIRE

No.	<i>Environmental Consciousness Questions</i>
1.	We are approaching the limit of the number of people the earth can support.
2.	Humans have the right to modify the natural environment to suit their needs.
3.	When humans interfere with nature it often produces disastrous consequences.
4.	Human ingenuity will insure that we do NOT make the earth unlivable.
5.	Humans are severely abusing the environment.
6.	The earth has plenty of natural resources if we just learn how to develop them.
7.	Plants and animals have as much right as humans to exist.
8.	The balance of nature is strong enough to cope with the impacts of modern industrial nations.
9.	Despite our social abilities humans are still subject to the laws of nature.
10.	The so-called "ecological crisis" facing humankind has been greatly exaggerated.
11.	The earth is like a spaceship with a very limited room and resources.
12.	Humans were meant to rule over the rest of nature.
13.	The balance of nature is very delicate and easily upset.
14.	Humans will eventually learn enough about how nature works to be able to control it.
15.	If things continue on their present course, we will soon experience a major ecological catastrophe.

Table 7: The Environmental Consciousness questionnaire, also called the New Ecological Paradigm (NEP) Scale, based upon the original work of Dunlap et al. [8]. The questions are graded on a 5-point Likert scale, with answers ranging from strongly disagree (1) to strongly agree (5). Question wording: "Below are some statements about the relationship between humans and the environment. For each statement, please indicate the extent to which you agree with it."

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